Mycelium: A Verification-First Spiking Network from the CeliumNeUR Constraint Set

Mario Gutiérrez
Celiums Solutions LLC
mario@celiums.ai

August 2026 — working draft v0.2

Abstract

Most spiking neural network (SNN) papers propose neuron dynamics and assert hardware efficiency conditionally, against silicon that does not exist. We invert the direction: we take CeliumNeUR — a verification-first neuromorphic SoC with synthesizable RTL, a bit-exact golden model, mutation testing and bounded formal checks — and derive a trainable model *from its constraint set*: ceiling leak, saturating 16-bit membranes, ternary saturating weights, per-neuron thresholds, leak and refractory periods, tick-synchronous phase semantics, and static routing. The result, **Mycelium**, is a recurrent block-sparse ternary architecture whose forward pass is bit-identical to the chip's verified golden model under an explicit five-condition contract, whose learned topology is frozen into a hashable, input-independent artifact, and whose deployment instance — 34 neurons and 832 synapse entries, inside the v1 silicon budget — replays through the integer reference with zero mismatches after training on real data. On Spiking Heidelberg Digits, Mycelium beats a budget-matched fp16 GRU on the memory frontier in the edge regime (62.3% ± 3.0 at 31 KB versus 55.4% ± 4.0 at 68 KB, three seeds each), where the per-operation frontier also tilts in its favor; dense wins above ∼300 KB and in raw operation count elsewhere — exactly the split a memory-bandwidth thesis predicts. Along the way, L_0 topology learning *deleted the recurrent synapse entirely* when the task did not pay for it; a float-weight ablation *lost* to ternary under identical dynamics; and a membrane-potential readout, chip-honest through the RTL's non-invasive readback port, improved both accuracy (+2.6 points) and seed stability (±1.9 vs. ±2.9) over a spike-count head.

1 Introduction: constraints first, hardware later

BitNet's ternary weights were motivated by multiplication-free hardware that GPUs lack; BitNet wins anyway — on memory bandwidth [6]. We claim the same structure holds for neuromorphic constraint sets. Split the chip's efficiency story in two:

- a **static half** — sparse frozen topology and integer weights — which is a *memory* property and transfers to commodity accelerators today;
- a **dynamic half** — event-driven "no spike, no work" execution — which is a *compute* property, worthless below a block granularity of ∼64 on GPUs, and latent until silicon exists.

Designing against the full constraint set from the first training step matters: a model trained with relaxed float dynamics (exponential leak, long subthreshold memory) and quantized afterward depends on memory the silicon does not have, and fails on deployment.

A second claim is unique to this project: **verifiability as a feature**. A frozen compute graph is an artifact — hashable, signable, publishable. Because CeliumNeUR ships a golden model that its RTL is verified against cycle-by-cycle (mutation testing: 17/17 killed; bounded model checking to depth 60; lint-clean synthesis receipts), a network that is bit-exact against that golden model inherits a chain of custody no SNN paper we know of can offer:

$$\text{task-trained weights} \;\equiv\; \text{integer model} \;\equiv\; \text{golden referee} \;\equiv\; \text{verified RTL.}$$

2 The constraint set

CeliumNeUR v1 is a 4-tile, 1,024-neuron, 1,024-synapse-entry SoC (2×2 mesh, X-first dimension-order routing, credit-based flow control). The authority order for semantics is: golden model > specification prose. The neuron, extracted from the golden source:

State. Membrane $v \in [-32768, 32767]$ (signed 16-bit, *saturating*, never wrapping); refractory countdown $c \geq 0$. Per-neuron parameters (all learnable in Mycelium, all stored per neuron on chip): threshold $\theta \in [1, 32767]$, leak shift $k \in [0, 15]$, refractory duration R, reset mode.

One global tick. With phase input sum I_t (spikes staged at tick $t-1$, integer counts × integer weights) and event-presence mask E_t:

$$v' = \mathrm{sat}_{16}(v_{t-1} + I_t) \tag{1}$$

$$f_t^{\mathrm{evt}} = E_t \wedge (c_{t-1} = 0) \wedge (v' \geq \theta) \tag{2}$$

$$v'' = v' - f_t^{\mathrm{evt}}\,\theta, \qquad c' = \begin{cases} R & f_t^{\mathrm{evt}} \\ c_{t-1} & \text{else} \end{cases} \tag{3}$$

$$v''' = \mathrm{sat}_{16}\big(v'' - \mathrm{sign}(v'')\,\left\lceil |v''|/2^k \right\rceil\big) \tag{4}$$

$$f_t^{\mathrm{tick}} = (c' = 0) \wedge (v''' \geq \theta) \tag{5}$$

$$v_t = v''' - f_t^{\mathrm{tick}}\,\theta, \qquad c_t = \max\big(c'[f_t^{\mathrm{tick}} \mapsto R] - 1,\ 0\big) \tag{6}$$

The ceiling leak loses at least one unit per tick: subthreshold memory is *proportional to signal magnitude*, never exponential-tailed. The event gate E_t is load-bearing (a zero-weight synapse still evaluates on chip; a quiet phase evaluates only post-leak), and the decrement-after-evaluation order makes the two fire paths deliberately asymmetric (an event-path fire blocks R ticks; a tick-path fire blocks $R-1$) — both discovered during extraction and since documented upstream as contract.

The equivalence contract (C1–C5). The golden model evaluates fire per synaptic event; a batched model evaluates per tick. Exact end-of-phase state equality holds under: **C1** subtractive reset, **C2** $R \geq 1$, **C3** no intermediate 16-bit saturation, **C4** no order-dependent threshold crossing, **C5** external injection subthreshold at injection time. Every boundary is pinned by an explicit divergence test. Under the contract we verified exact spike and state equality on over 10^5 fuzzed neuron-phases and on full 1,024-neuron chip simulations (mesh, dendrite tables with true duplicate multiplicity, somas), with delivery routed through the same sparse primitive used for training.

3 Mycelium

Macro-architecture. One or more recurrent modules over a *static block graph*: feed and recurrent synapses are block-sparse ternary matrices (an active $B \times B$ block is all B^2 dendrite entries, making event semantics exact and the mask a faithful FLOP accountant), neurons are the integer dynamics above with a straight-through/surrogate training path, and a linear head reads the top layer's *membrane potential in threshold units* v/θ at every tick, averaged. The membrane head is chip-honest: the RTL exposes soma state through a non-invasive readback port (invariant I5), so the head is a host-side layer; a spike-count head is the fully-on-chip alternative at -2.6 points and $1.5\times$ the seed variance. No attention, no data-dependent routing: the compute graph is input-independent and hashable.

Training. Surrogate gradients (arctan kernel, width $0.25\,\theta$) at both comparators; straight-through estimators through the ceiling leak, weight rounding, and the threshold grid; hard saturation passes no gradient beyond the rails. Weights are ternarized per block by absmean [6]; thresholds are float masters STE-rounded to the chip grid; leak is heterogeneous per neuron across the full 4-bit range [8]. Topology is learned with hard-concrete L_0 gates [7] over the block grid at a hot learning rate, then *binarized and frozen*; fine-tuning re-accommodates weights to the fixed graph, whose masks hash to a SHA-256 artifact. Adam with cosine decay; spike dropout; no gradient clipping (it hurt at every strength we tried).

4 Results

All numbers reproducible from the repository's `experiments/` JSONs.

4.1 Gradient viability (pre-registered gate)

A 324-configuration sweep (surrogate shape $\times$ width $\times$ weight precision $\times$ T $\times$ leak $\times$ threshold placement) passed every pre-registered kill criterion. A previously observed null result ($\sim 2 \times 10^{-5}$ gate gradient in a ternary predecessor) reproduced *only* in the sweep's worst corner — fast leak, high threshold, compact-support (triangular) kernel — identifying it as a threshold-placement pathology rather than a ternarization cost. Ternary was the *best*-conditioned precision (median relative input-layer gradient 3.1×10^{-2}, vs. float 6.8×10^{-3} and int8 2.4×10^{-4}); fat-tailed kernels are robust where triangular dies; slow leak dominates trainability; BPTT carries credit at $T{=}32$ without vanishing.

4.2 Topology learning

On a task with a planted-relevance structure (one signal input block, one pure-noise block), L_0 gating closed every noise route exactly at zero accuracy cost, and random masks at the learned density collapsed to chance half the time. On SHD, the learner made an unprompted structural decision: at $\lambda = 0.15$ it **deleted the recurrent synapse entirely** (-1.9 points, $3\times$ fewer operations) — rate-readout SHD at $T{=}32$ does not pay for recurrence, and the method discovered that by itself.

4.3 Task curves

Spiking Heidelberg Digits [1] (20 classes, 700 spike channels, integer count binning at $T{=}32$), against fp16 GRUs trained under the same budget (60 epochs cosine, learning-rate swept). **Every row: three seeds.**

Model	Accuracy	Bytes	MMAC/sample
Mycelium λ=0.15	$\mathbf{0.623 \pm 0.030}$	**31 KB**	**1.5**
GRU-16 (fp16)	0.554 ± 0.040	68 KB	1.1
GRU-32 (fp16)	0.669 ± 0.033	140 KB	2.3
Mycelium λ=0.02	0.673 ± 0.019	167 KB	19.3
Mycelium H=1024	0.650 ± 0.021	284 KB	32.1
GRU-256 (fp16, T=64+aug)	0.883 ± 0.007	1453 KB	47.5

Table 1: SHD frontier. Bytes: 2-bit ternary weights in active blocks + block masks + per-neuron soma parameters vs. fp16 dense. MACs count ternary additions as multiplies and ignore ∼0.4 hidden event sparsity — both conservative against Mycelium.

The byte frontier favors Mycelium below ∼150 **KB** (+6.9 points at less than half the bytes at the extreme edge; a tie near 150 KB) and dense above ∼300 KB. **The per-operation frontier crosses in the extreme-edge corner**: at 1.5 MMAC Mycelium scores 0.623 where the dense interpolation between its bracketing points passes ≈0.59; elsewhere dense wins per-op. Doubling temporal resolution and adding augmentation moved the GRU +5 points and Mycelium none: the remaining accuracy gap at scale is architectural, of which the membrane head recovered +2.6 points while halving seed variance. Multi-seeding also showed the small dense baselines to be equally noisy (GRU-16: ±4.0): at this scale, seed sensitivity is a property of the regime, not of spiking. DVS128 Gesture [9] (11 classes, chance 9.1%) replicates the pattern with the spike head: 0.689 at 133 KB vs. GRU 0.773 at 484 KB.

4.4 The deployment certificate

A two-class SHD instance inside the v1 silicon budget — 34 neurons and 832 dendrite entries, including the host-side encoder — trains to 74.9% and replays through the integer model (bit-exact to the golden referee, hence to RTL semantics) with **zero logit mismatches** over the full test set. The artifact is a SHA-256 over edges, int8 weights and integer thresholds. To our knowledge this is the first task-trained SNN with a machine-checkable equivalence chain to silicon-verified RTL.

5 Related work

ODIN [2] and ReckOn [3] define the digital-neuromorphic baseline CeliumNeUR's own specification calibrates against. snnTorch [4] supplies the surrogate-gradient canon; we deliberately did not fork it — a tested adapter exposes our neuron through its interface instead. Spikformer [5] shows spiking attention exists, but attention is data-dependent routing — precisely what a signable static graph excludes. BitNet b1.58 [6] is the structural precedent for the memory thesis; hard-concrete L_0 [7] provides the gate machinery; neural heterogeneity [8] motivates per-neuron time constants, which the chip stores natively. Cramer et al. [1] report recurrent-LIF baselines around 0.71 at higher T, situating our 0.62–0.67 at T=32 as in-family for spiking models, not state of the art. Surrogate-gradient learning follows [10, 11]; the arctan kernel is from [12].

6 Limitations

Inference-side efficiency only (BPTT training costs more than dense); byte accounting at 2 bits/weight (1.58 information-theoretic would shift the Mycelium curve further left); one and a half tasks, with the membrane head unevaluated on DVS; three seeds per point (honest error bars, not significance theater — the edge-point gap is $\sim 2\sigma$); no GPU block-sparse kernel wall-clock measurements (FLOP fractions stand in); the chip-faithful instance is a deployability certificate, not a competitive model; scaling past 1,024 neurons on-silicon requires a flit redesign.

7 Conclusion

Constraint-first design paid where the thesis said it would: memory. The verification chain — model to silicon referee, bit-exact, signed — is the claim we recommend leading with; the edge-regime frontier win is the number that makes it more than a curiosity.

Artifacts. Model: `github.com/terrizoaguimor/celiumsnn` (Apache-2.0). Chip: `github.com/terrizoaguimo` (Apache-2.0), DOI `10.5281/zenodo.21925426`.

References

[1] B. Cramer, Y. Stradmann, J. Schemmel, F. Zenke. The Heidelberg spiking data sets for the systematic evaluation of spiking neural networks. *IEEE TNNLS*, 2020.

[2] C. Frenkel, M. Lefebvre, J.-D. Legat, D. Bol. A 0.086-mm^2 12.7-pJ/SOP 64k-synapse 256-neuron online-learning digital spiking neuromorphic processor in 28-nm CMOS. *IEEE TBioCAS*, 13(1), 2019.

[3] C. Frenkel, G. Indiveri. ReckOn: A 28nm sub-mm^2 task-agnostic spiking recurrent neural network processor enabling on-chip learning over second-long timescales. *ISSCC*, 2022.

[4] J. K. Eshraghian et al. Training spiking neural networks using lessons from deep learning. *Proc. IEEE*, 111(9), 2023.

[5] Z. Zhou et al. Spikformer: When spiking neural network meets transformer. *ICLR*, 2023.

[6] S. Ma et al. The era of 1-bit LLMs: All large language models are in 1.58 bits. arXiv:2402.17764, 2024.

[7] C. Louizos, M. Welling, D. P. Kingma. Learning sparse neural networks through L_0 regularization. *ICLR*, 2018.

[8] N. Perez-Nieves, V. C. H. Leung, P. L. Dragotti, D. F. M. Goodman. Neural heterogeneity promotes robust learning. *Nature Communications*, 12:5791, 2021.

[9] A. Amir et al. A low power, fully event-based gesture recognition system. *CVPR*, 2017.

[10] E. O. Neftci, H. Mostafa, F. Zenke. Surrogate gradient learning in spiking neural networks. *IEEE Signal Processing Magazine*, 36(6), 2019.

[11] F. Zenke, T. P. Vogels. The remarkable robustness of surrogate gradient learning for instilling complex function in spiking neural networks. *Neural Computation*, 33(4), 2021.

[12] W. Fang et al. Incorporating learnable membrane time constant to enhance learning of spiking neural networks. *ICCV*, 2021.